Highlights

- Four top-taggers (BDT, CNN, ParticleNet, Particle Transformer) compared under one pipeline.
- Data-efficiency sweep reveals a BDT-to-transformer crossover as training data grows.
- Ranking and calibration are decoupled: the CNN is worst-calibrated despite mid ranking.
- Temperature scaling fixes all models except the structurally miscalibrated CNN.
- Fully reproducible on a single free cloud GPU; code and models released openly.

Data efficiency and probability calibration of deep jet-tagging architectures on the 14 TeV top-quark benchmark

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Abstract

Top tagging — identifying the hadronic decays of highly boosted top quarks — is a standard benchmark for machine learning in collider physics. We compare four classifiers on the public Top-Quark Tagging Reference Dataset of 14 TeV proton–proton collisions: a gradient-boosted decision tree (BDT) on engineered jet-substructure variables, a convolutional neural network (CNN) on jet images, the ParticleNet graph network, and the Particle Transformer. Rather than chasing a new record area under the curve (AUC), we study two properties of direct relevance to downstream analyses: *data efficiency* and *probability calibration*. Using a fully reproducible pipeline trained on a single free cloud GPU, we sweep the training-set size from 5×10^3 to 5×10^4 jets and measure test AUC, background rejection at fixed signal efficiency, the Expected Calibration Error (ECE), and the effect of temperature scaling. The Particle Transformer attains the best discrimination (AUC = 0.978), reproducing the published ordering, but the data-efficiency sweep reveals a clear crossover: the BDT is strongest at the smallest training size and is only overtaken by the deep models as data grows. The calibration study shows that ranking power and calibration are decoupled — the jet-image CNN is the worst-calibrated model (ECE = 0.078) despite a mid-pack ranking, whereas the top-ranked transformer is comparatively well-calibrated, and a single-parameter temperature rescaling corrects every model except the CNN. The complete pipeline, trained models and documentation are released as open

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source.

Keywords: top tagging, jet substructure, deep learning, ParticleNet, Particle Transformer, data efficiency, probability calibration

1. Introduction

1.1. Physics context

The top quark is the heaviest known elementary particle, with a mass $m_t \simeq 172.76\,\mathrm{GeV}$ [1]. Because this mass exceeds the hadronisation scale, the top decays as a bare quark, almost always through $t \to Wb$; in the fully hadronic channel the W decays as $W \to q\bar{q}'$, giving $t \to Wb \to q\bar{q}'b$. When the parent top carries a transverse momentum $p_T \gg m_t$ it is *boosted*: its three decay quarks are produced collinearly and, after parton showering and hadronisation, merge into one large-radius jet reconstructed by sequential-recombination algorithms such as anti-k_t [2, 3, 4]. Such a "top jet" differs from an ordinary jet initiated by a light quark or gluon (a QCD jet) in three measurable ways: it is more massive (mass $\sim m_t$), it shows a three-pronged rather than one-pronged radiation pattern, and it carries a higher constituent multiplicity. Exploiting these differences to separate top jets from the overwhelmingly more abundant QCD background — *top tagging* — is one of the canonical classification problems of LHC physics and a central tool in searches for physics beyond the Standard Model [5, 6].

1.2. From engineered features to deep learning

Classical taggers compress the jet into a handful of physically motivated *substructure observables*: the jet mass, the N-subjettiness ratios τ_{21}, τ_{32} [7], and energy-correlation functions such as C_2 and D_2 [8, 9]. Fed to a boosted decision tree (BDT), these provide a strong, interpretable baseline. The deep-learning era replaced hand-crafted features with representations learned directly from the constituents. Three paradigms followed in succession: pixelating the jet into a calorimeter-like image for a convolutional neural network (CNN) [10, 11]; treating the jet as an unordered "particle cloud" processed by a permutation-invariant graph network such as ParticleNet [12], built on the Dynamic Graph CNN [13]; and, most recently, modelling pairwise constituent interactions with self-attention in the Particle Transformer [14, 15]. Related set, graph and Lorentz-equivariant architectures continue to advance the field

[16, 17, 18, 19]. On the standard reference dataset [20], a community comparison [21] established the now-canonical ordering, with the leading deep models exceeding an area under the receiver operating characteristic curve (AUC) of 0.98.

1.3. Motivation and contributions

The published literature is almost entirely concerned with the *best achievable* discrimination, quoted at the full training-set size of 1.2×10^6 jets and on large GPU clusters. Two properties of direct practical importance are rarely studied together. The first is *data efficiency*: generating labelled Monte-Carlo training samples is computationally expensive, so how steeply each architecture's performance depends on the amount of training data — and whether the ranking of the models changes in the small-data regime — is decisive for groups without large simulation budgets. The second is *probability calibration*: a high AUC certifies only that a model *ranks* signal above background, whereas many downstream analyses use the classifier score as a *probability* (for instance in a profile-likelihood fit), which requires the score to be calibrated. While calibration is routinely examined in the machine-learning literature [22, 23], it is seldom reported for jet taggers.

This paper addresses both questions on the public reference dataset using a single, reproducible pipeline that runs end-to-end on free hardware. Our contributions are:

1. a unified, open-source pipeline that trains and evaluates the BDT, CNN, ParticleNet and Particle Transformer under identical preprocessing, data splits and random seeds;
2. a data-efficiency study across a decade of training-set sizes (5×10^3 to 5×10^4 jets), revealing a clear crossover between the BDT and the deep models; and
3. a probability-calibration study, using reliability diagrams, the Expected Calibration Error and temperature scaling, showing that ranking power and calibration are decoupled, with practical recommendations for downstream use.

The remainder of the paper is organised as follows. Section 2 describes the dataset and preprocessing; Section 3 sets out the four models and the evaluation methodology with their governing equations; Section 4 presents the headline benchmark, the data-efficiency sweep and the calibration analysis; and Section 5 concludes.

2. Dataset and preprocessing

We use the public Top-Quark Tagging Reference Dataset [20, 21], 2×10^6 jets from 14 TeV pp collisions generated with Pythia 8 [24] and a Delphes fast detector simulation [25], clustered with the anti-k_t algorithm [2, 3] at $R = 0.8$ and $550 < p_T < 650$ GeV. For tractability we draw stratified subsamples of up to 5×10^4 jets per split and use the leading 100 constituents per jet.

For each constituent we form the feature vector $(p_T, \eta, \phi, E, \Delta\eta, \Delta\phi, \log p_T)$ relative to the p_T-weighted jet axis, standardised on the training split. For the BDT we compute thirteen engineered observables (jet mass, p_T, η, multiplicity, width, leading/sub-leading p_T fractions, $\tau_1, \tau_2, \tau_3, \tau_{21}, \tau_{32}$ [7], and C_2 [8, 9]); their distributions (Fig. 1) behave exactly as physics expects — the top mass peaks at m_t and τ_{32} separates the three-prong signal. For the CNN we bin constituents into $40 \times 40 \times 3$ jet images [10]; the averaged images (Fig. 2) show the three-prong topology in the signal−background difference. For the Particle Transformer we add four pairwise features $(\ln \Delta R, \ln k_T, z, \Delta\eta)$ following Qu et al. [14].

3. Methodology

We treat top tagging as binary classification: each jet $\mathbf{x}$ is mapped by a model f_θ to a score $\hat{y} = f_\theta(\mathbf{x}) \in [0, 1]$ estimating the probability that the jet is a top jet. All four models are trained against the binary cross-entropy loss

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^{N} \Big[y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i) \Big], \tag{1}$$

where $y_i \in \{0, 1\}$ is the truth label and the neural models produce $\hat{y}_i = \sigma(z_i)$ from a logit z_i through the logistic sigmoid $\sigma(z) = 1/(1 + e^{-z})$.

3.1. Boosted decision tree (BDT)

The baseline is a gradient-boosted decision-tree ensemble trained with XGBoost [26] on the thirteen engineered observables. The prediction is an additive ensemble of K regression trees $f_k \in \mathcal{F}$,

$$z_i = \sum_{k=1}^{K} f_k(\mathbf{x}_i), \tag{2}$$

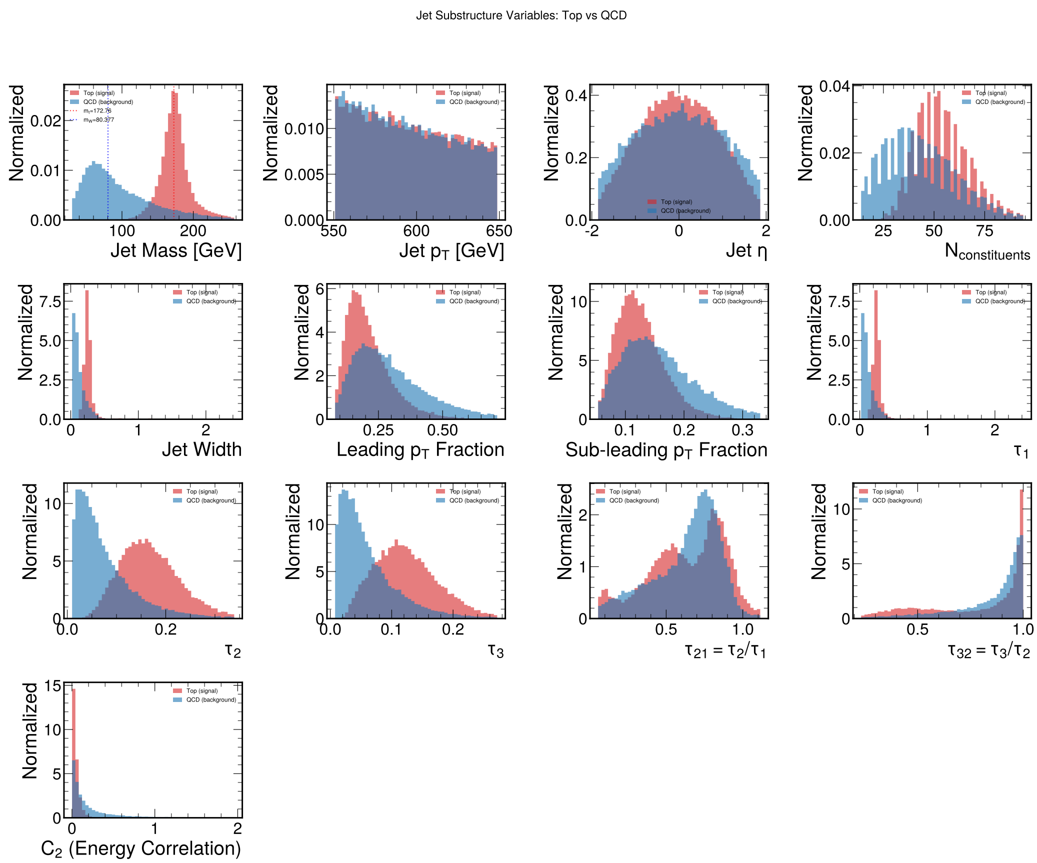

Figure 1: Engineered jet-substructure observables for top (red) and QCD (blue) jets on the test set.

built greedily by minimising the regularised objective

$$\mathcal{O} = \sum_i \ell(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \qquad \Omega(f) = \gamma T + \tfrac{1}{2}\lambda\|\mathbf{w}\|^2, \tag{3}$$

where ℓ is the logistic loss, T is the number of leaves, $\mathbf{w}$ the leaf weights, and γ, λ control complexity. At each boosting round a new tree is fit to the second-order Taylor expansion of $\mathcal{O}$, which yields fast, accurate splits. Among the inputs, the N-subjettiness ratios [7]

$$\tau_N = \frac{1}{d_0} \sum_k p_{T,k} \min\bigl(\Delta R_{1,k}, \ldots, \Delta R_{N,k}\bigr), \qquad \tau_{21} = \frac{\tau_2}{\tau_1}, \quad \tau_{32} = \frac{\tau_3}{\tau_2}, \tag{4}$$

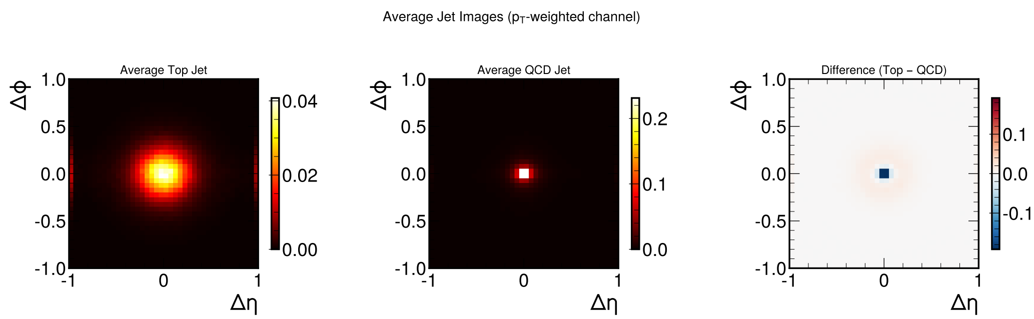

Figure 2: Averaged p_T-weighted jet images: top (left), QCD (centre) and their difference (right).

($d_0 = \sum_k p_{T,k} R_0$, $\Delta R_{a,k}$ the angular distance from constituent k to axis a) measure compatibility with an N-pronged hypothesis and, together with the jet mass and the jet width, rank among the most important inputs (Section 4.4).

3.2. Convolutional neural network (CNN)

The CNN acts on the $40 \times 40 \times 3$ jet image I, applying learnable kernels W in the $(\Delta\eta, \Delta\phi)$ plane,

$$(I * W)_{ij}^{c'} = \sum_c \sum_{a,b} W_{a,b}^{c',c} \, I_{i+a,\, j+b}^{c}, \tag{5}$$

interleaved with batch normalisation and ReLU nonlinearities $\mathrm{ReLU}(u) = \max(0, u)$. We use a ResNet-style stack [27] in which each block learns a residual mapping

$$\mathbf{h}_{\ell+1} = \mathrm{ReLU}\big(\mathbf{h}_\ell + \mathcal{F}(\mathbf{h}_\ell; W_\ell)\big), \tag{6}$$

easing optimisation of deeper networks. Global average pooling and a fully connected head produce the logit z.

3.3. ParticleNet (graph network)

ParticleNet [12] treats the jet as an unordered point cloud $\{\mathbf{x}_i\}$ and is built from EdgeConv blocks [13]. For each particle i, a k-nearest-neighbour graph is constructed in coordinate space and the feature is updated by a symmetric aggregation over its neighbourhood $\mathcal{N}(i)$,

$$\mathbf{x}_i' = \max_{j \in \mathcal{N}(i)} h_\Theta\big(\mathbf{x}_i,\ \mathbf{x}_j - \mathbf{x}_i\big), \tag{7}$$

where h_Θ is a shared multilayer perceptron and the element-wise max makes the operation permutation-invariant. The graph is recomputed in the learned feature space at each block (a "dynamic" graph), so successive layers capture progressively larger-scale structure; a final global pooling and dense head give z.

3.4. Particle Transformer

The Particle Transformer [14] applies multi-head self-attention [15] to the constituent sequence. With queries, keys and values Q, K, V obtained by linear projection, a standard attention head computes

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \tag{8}$$

with d_k the key dimension. The key physics ingredient is a learned *interaction bias*: the four pairwise features $\mathbf{u}_{ij} = (\ln \Delta R_{ij}, \ln k_{T,ij}, z_{ij}, \Delta\eta_{ij})$ are mapped by an MLP to a per-head additive term $U_{ij} = \phi(\mathbf{u}_{ij})$ that is injected into the attention logits,

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + U\right) V, \tag{9}$$

so that physically close, hard splittings are emphasised. A learnable class token aggregates the sequence and its representation is mapped to the logit z; its attention weights provide an interpretability handle (Section 4).

3.5. Training

The neural models minimise Eq. (1) with Adam/AdamW [28, 29] (learning rates 10^{-3}–10^{-4}), gradient clipping, a plateau learning-rate scheduler and early stopping on the validation AUC, implemented in PyTorch [30] with NumPy [31], scikit-learn [32] and Matplotlib [33]. Training used a free Google Colaboratory T4 GPU; the pipeline checkpoints every epoch and resumes automatically.

3.6. Evaluation metrics

We report the AUC, the accuracy at a score threshold of 0.5, and the background rejection $1/\varepsilon_B$ at fixed signal efficiencies $\varepsilon_S = 0.5, 0.3$, the figure of merit most relevant to physics analyses; AUC uncertainties are obtained by bootstrap resampling of the test set. Calibration is quantified by the Expected Calibration Error [22], which partitions the predictions into M

equal-width score bins B_m and averages the gap between confidence and accuracy,

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left| \mathrm{acc}(B_m) - \mathrm{conf}(B_m) \right|. \tag{10}$$

As a corrective we apply *temperature scaling*: a single scalar $T > 0$ is fit on the validation set by minimising the negative log-likelihood of the rescaled scores $\hat{y} = \sigma(z/T)$, and we report the ECE before and after rescaling. $T > 1$ indicates an over-confident model whose probabilities are softened by the rescaling.

4. Results and Discussion

4.1. Headline benchmark

Table 1 and Fig. 3 report the test-set performance of the four models trained on 5×10^4 jets. The ordering Particle Transformer (AUC = 0.978) > ParticleNet (0.976) > CNN (0.972) > BDT (0.970) reproduces the community benchmark [21]. The same ordering is even sharper in the background-rejection metric, which weights the high-purity regime most relevant to analyses: at a signal efficiency $\varepsilon_S = 0.5$ the rejection rises from $1/\varepsilon_B = 93$ (BDT) to 136 (Particle Transformer), and at $\varepsilon_S = 0.3$ from 287 to 476 — a $\sim 1.6\times$ improvement of the attention model over the engineered- feature baseline. The score distributions (Fig. 4) are strongly bimodal, peaking near 0 and 1, and the separation visibly sharpens from the BDT to the deep models; the confusion matrices (Fig. 5) at a threshold of 0.5 confirm balanced signal/background accuracies of 91–92%.

Our absolute AUCs sit a little below the published full-dataset values (ParticleNet and the Particle Transformer reach $\sim$0.985 and $\sim$0.986 respectively when trained on the full 1.2×10^6 jets with larger models and longer schedules [12, 14]). This is expected: we deliberately use a 5×10^4-jet subset and compute-trimmed configurations so the entire study runs on a single free GPU. Crucially, the *relative* ordering and the gaps between models are preserved, which is what the data-efficiency and calibration analyses below build upon.

4.2. Data efficiency

Re-training each model from scratch at $N_{\mathrm{train}} \in \{5, 10, 25, 50\} \times 10^3$ jets yields the learning curves of Fig. 6 and Table 2. Three distinct behaviours

Table 1: Headline test-set performance (5×10^4-jet training run). $1/\varepsilon_B^{0.5}$ and $1/\varepsilon_B^{0.3}$ are background rejections at signal efficiencies 0.5 and 0.3.

Model	AUC	Acc.	$1/\varepsilon_B^{0.5}$	$1/\varepsilon_B^{0.3}$
BDT (XGBoost)	0.970	0.911	93	287
CNN (jet images)	0.972	0.915	111	413
ParticleNet	0.976	0.920	130	413
Particle Transformer	0.978	0.925	136	476

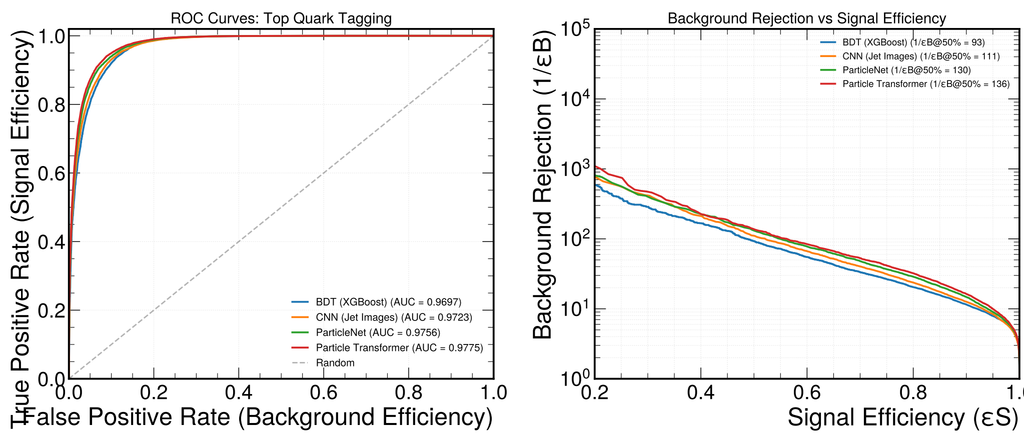

Figure 3: ROC curves (left) and background rejection $1/\varepsilon_B$ vs. signal efficiency ε_S (right) for the four models on the test set.

emerge. The BDT is the most *data-efficient* model: its AUC rises by less than one percent ($0.961 \to 0.968$) across the whole decade of training sizes, because its inputs are physics-engineered observables (Eq. 4) that already encode a strong inductive bias, leaving little for the learner to discover from data. The deep models behave oppositely — they must learn their representations from the raw constituents and therefore improve steeply with N_{train} — which produces a clear *crossover*: the Particle Transformer is the *weakest* model at 5×10^3 jets (0.949) but the *strongest* at 5×10^4 (0.974), overtaking both the BDT and ParticleNet around $N_{\text{train}} \approx 2.5 \times 10^4$. The CNN is intermediate: it benefits from data but plateaus earlier than the permutation-equivariant models, consistent with the information lost in the pixelisation step. The background-rejection panel of Fig. 6 tells the same story with a steeper slope, and the bootstrap uncertainties in Table 2 (± 0.001) confirm that the crossover is statistically significant rather than a fluctuation.

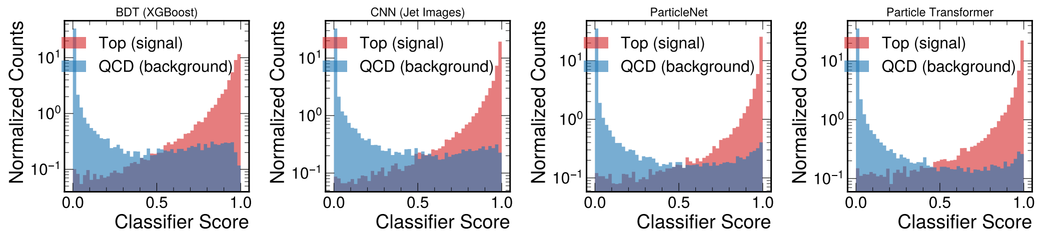

Figure 4: Classifier-score distributions for top (red) and QCD (blue) jets. The bimodal peaks near 0 and 1 sharpen from the BDT to the deep models.

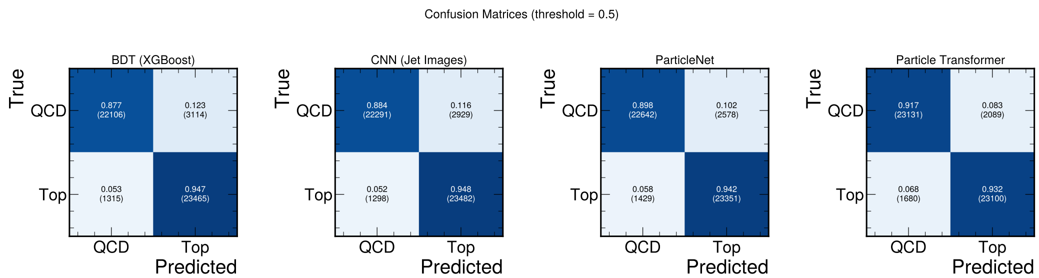

Figure 5: Normalised confusion matrices at a decision threshold of 0.5.

The practical implication is direct: under a tight Monte-Carlo budget the simple, interpretable BDT is competitive with — or preferable to — the deep models, whereas the attention model only realises its advantage once $\gtrsim 10^4$ jets are available, in line with its reported near-10^6- jet saturation [14].

4.3. Probability calibration

Figure 7 and Table 3 report the calibration at the 5×10^4-jet endpoint. All four models are mildly over-confident (every fitted temperature satisfies $T > 1$), as is typical of modern classifiers [22], but to very different degrees. Strikingly, ranking power and calibration are *decoupled*. The jet-image CNN is the worst-calibrated model by an order of magnitude (ECE $= 0.078$, $T = 1.54$) despite ranking third on AUC, while the top-ranked Particle Transformer is comparatively well-calibrated (ECE $= 0.012$) and the BDT and ParticleNet are already excellent (ECE ≈ 0.007). The reliability diagram for the CNN sags well below the diagonal across the mid-score range, the signature of a model that reports probabilities systematically higher than the empirical positive rate.

We attribute the CNN's miscalibration to its representation: pixelisation

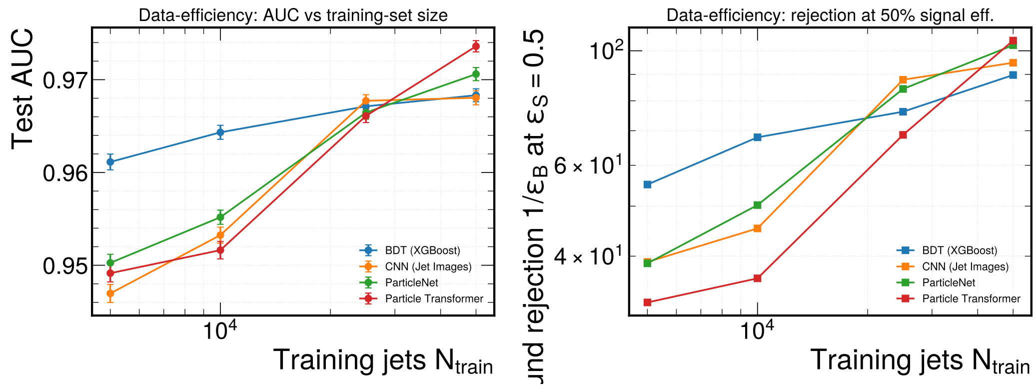

Figure 6: Data-efficiency curves: test AUC (left, with 1σ bootstrap error bars) and background rejection at $\varepsilon_S = 0.5$ (right) vs. training-set size. The Particle Transformer crosses from worst to best.

Table 2: Test AUC versus training-set size N_{train}. Uncertainties are one standard deviation from a 200-sample bootstrap of the test set.

Model	5×10^3	10^4	2.5×10^4	5×10^4
BDT (XGBoost)	0.961±0.001	0.964±0.001	0.967±0.001	0.968±0.001
CNN (jet images)	0.947±0.001	0.953±0.001	0.968±0.001	0.968±0.001
ParticleNet	0.950±0.001	0.955±0.001	0.966±0.001	0.971±0.001
Particle Transformer	0.949±0.001	0.952±0.001	0.966±0.001	0.974±0.001

discards angular information, after which the network fits a sharp decision boundary that piles scores towards 0 and 1 (visible in the bottom row of Fig. 7), inflating confidence without a matching gain in accuracy. Temperature scaling — a single global parameter — brings the BDT, ParticleNet and Particle Transformer to ECE $\lesssim 0.008$ essentially for free, but only halves the CNN's error (to 0.064): because the CNN's distortion is shape-dependent rather than a uniform over-confidence, a single temperature cannot remove it, and a richer non-parametric recalibration such as isotonic regression [23] would be required. For downstream analyses that treat the tagger score as a probability, this means temperature-scaled outputs are safe for three of the four models but the raw CNN score should not be used as a probability.

4.4. Interpretability

Two complementary views explain *why* the models discriminate. For the BDT, the gain-based feature importance (Fig. 8) is dominated by the jet

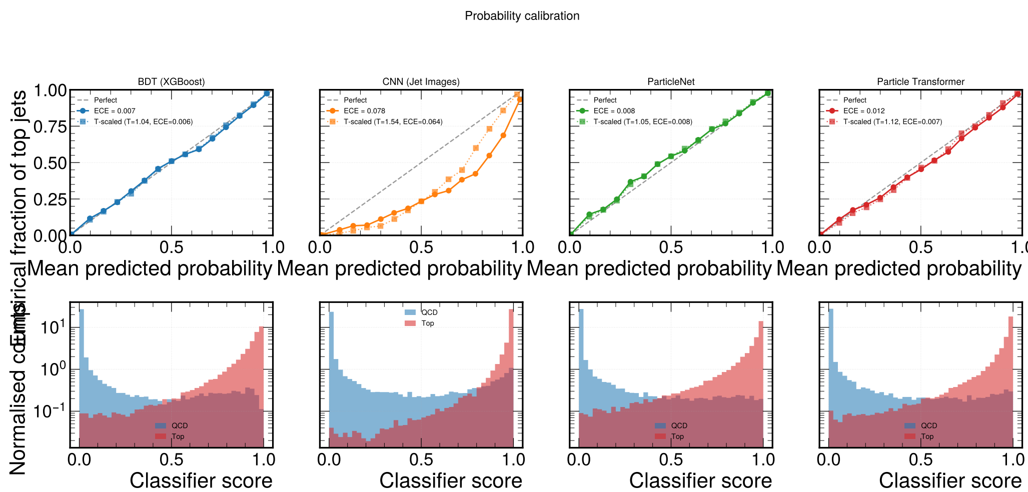

Figure 7: Reliability diagrams (top; solid: raw scores, dotted: temperature-scaled) and score histograms (bottom) for the four models. The jet-image CNN deviates most strongly from the diagonal.

Table 3: Calibration at the 5×10^4-jet endpoint: ECE, fitted temperature T, and ECE after temperature rescaling (ECE_T).

Model	ECE	T	ECE_T
BDT (XGBoost)	0.007	1.04	0.006
CNN (jet images)	0.078	1.54	0.064
ParticleNet	0.008	1.05	0.008
Particle Transformer	0.012	1.12	0.007

mass (gain 118), the 1-subjettiness τ_1 (89) and the jet width (76), with the constituent multiplicity next (15) — all observables that respond to the larger mass, girth and prong count of top jets, confirming that the tagger keys on physically meaningful structure rather than artefacts. Notably, the canonical ratio τ_{32} ranks lower (gain 7): the boosted trees capture most of the prong information through the *absolute* τ_1 and the jet width rather than through the ratio, even though τ_{32} is the cleaner discriminant at the distribution level (Fig. 1).

For the Particle Transformer, the learned class-token attention (Fig. 9) concentrates at small angular distance $\Delta R \lesssim 1$ for QCD jets, consistent with a single dominant prong, but spreads out to $\Delta R \sim 3$–4 for top jets, mirroring

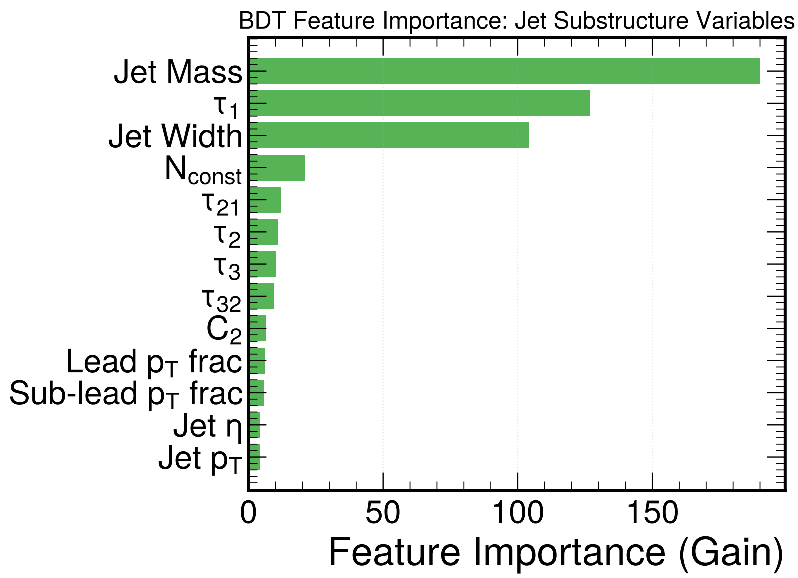

Figure 8: Gain-based BDT feature importance. The jet mass, τ_1 and jet width dominate, followed by the constituent multiplicity.

the broader, multi-pronged energy flow of the $t \to Wb \to q\bar{q}'b$ decay. The model has thus learned to attend to the peripheral radiation that carries the top signature. The per-layer self-attention maps (Fig. 10) show structured off-diagonal correlations among the hardest constituents, in contrast to the diffuse patterns of QCD jets, indicating that the interaction-bias term of Eq. (9) is being used to relate physically connected particles.

4.5. Discussion

Taken together, the three analyses give a more complete picture of these taggers than an AUC table alone. First, the headline ranking is robust but its *margins are budget-dependent*: at 5×10^4 jets the spread between the best and worst model is only $\sim$0.008 in AUC, and the data-efficiency sweep shows this spread opening up only as data grows. The crossover near $N_{\text{train}} \approx 2.5 \times 10^4$ quantifies a trade-off that is rarely stated explicitly: the value of a high-capacity model is conditional on the available simulation statistics. Second, the calibration results caution against equating ranking power with reliability — the CNN, third on AUC, is by far the least trustworthy as a probability, and this would directly bias a likelihood-based measurement that consumed its raw score. Third, the interpretability analysis confirms that both the

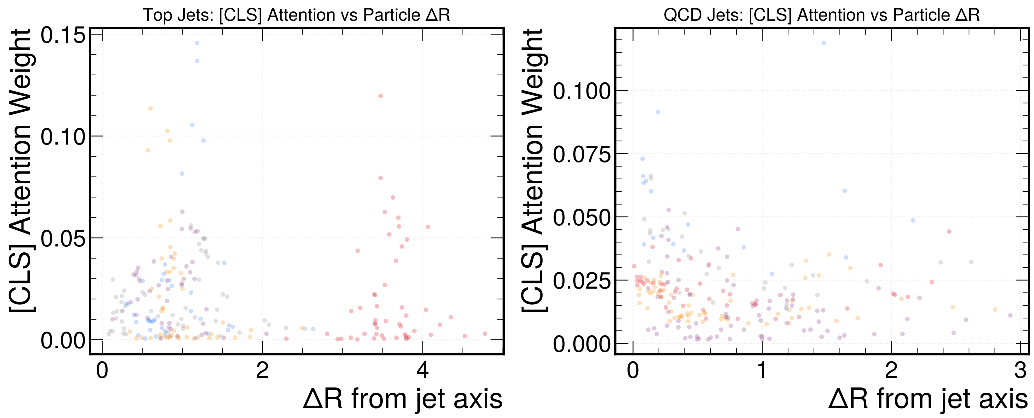

Figure 9: Class-token attention weight vs. constituent angular distance ΔR from the jet axis, for top jets (left) and QCD jets (right).

classical and the attention model exploit the same underlying physics (jet mass, subjettiness/width and multiplicity), which lends confidence that the learned discriminators are not artefacts of the simulation.

These findings are consistent with, and extend, the community comparison of Kasieczka et al. [21]: we reproduce its ordering on a far smaller budget and add the orthogonal axes of data efficiency and calibration. The main limitations are the reduced training-set size and model capacity (chosen for reproducibility on free hardware), the absence of pile-up and detector systematics in the reference sample, and the use of a single representative configuration per architecture rather than an exhaustive hyper-parameter scan. Each is a natural target for the follow-up work outlined in Section 5.

5. Conclusions

We have presented a reproducible comparison of four top-tagging architectures focused on data efficiency and probability calibration. The literature ordering is reproduced, but two practical messages emerge. First, data efficiency separates the models: the BDT is best in the small-data regime, with the deep models overtaking it only as data grows. Second, ranking and calibration are decoupled — the best-ranked transformer is well-calibrated, while the CNN, mid-pack on AUC, is the worst-calibrated and is not fully corrected by temperature scaling. We recommend reporting temperature-scaled probabilities by default and treating CNN scores with particular caution. Natural extensions are to scale the sweep to the full dataset, to study robustness under pile-up, and to compare non-parametric recalibration.

Code and data availability

The complete pipeline, trained models, and documents are openly available at `https://github.com/ramc77/fyp-jet-tagging`. The dataset is the public Top-Quark Tagging Reference Dataset [20] (`https://zenodo.org/records/2603256`).

CRediT authorship contribution statement

Bibi Naifa: Methodology, Software, Investigation, Writing – original draft. **Mah Lacka:** Software, Validation, Visualization, Writing – original draft. **Kanwal Gul:** Formal analysis, Data curation, Writing – review & editing. **Firdos:** Investigation, Validation, Writing – review & editing. **Ram Chand:** Conceptualization, Supervision, Project administration, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank the maintainers of the Top-Quark Tagging Reference Dataset and the open-source scientific-Python and Google Colaboratory ecosystems. This research received no specific grant from any funding agency.

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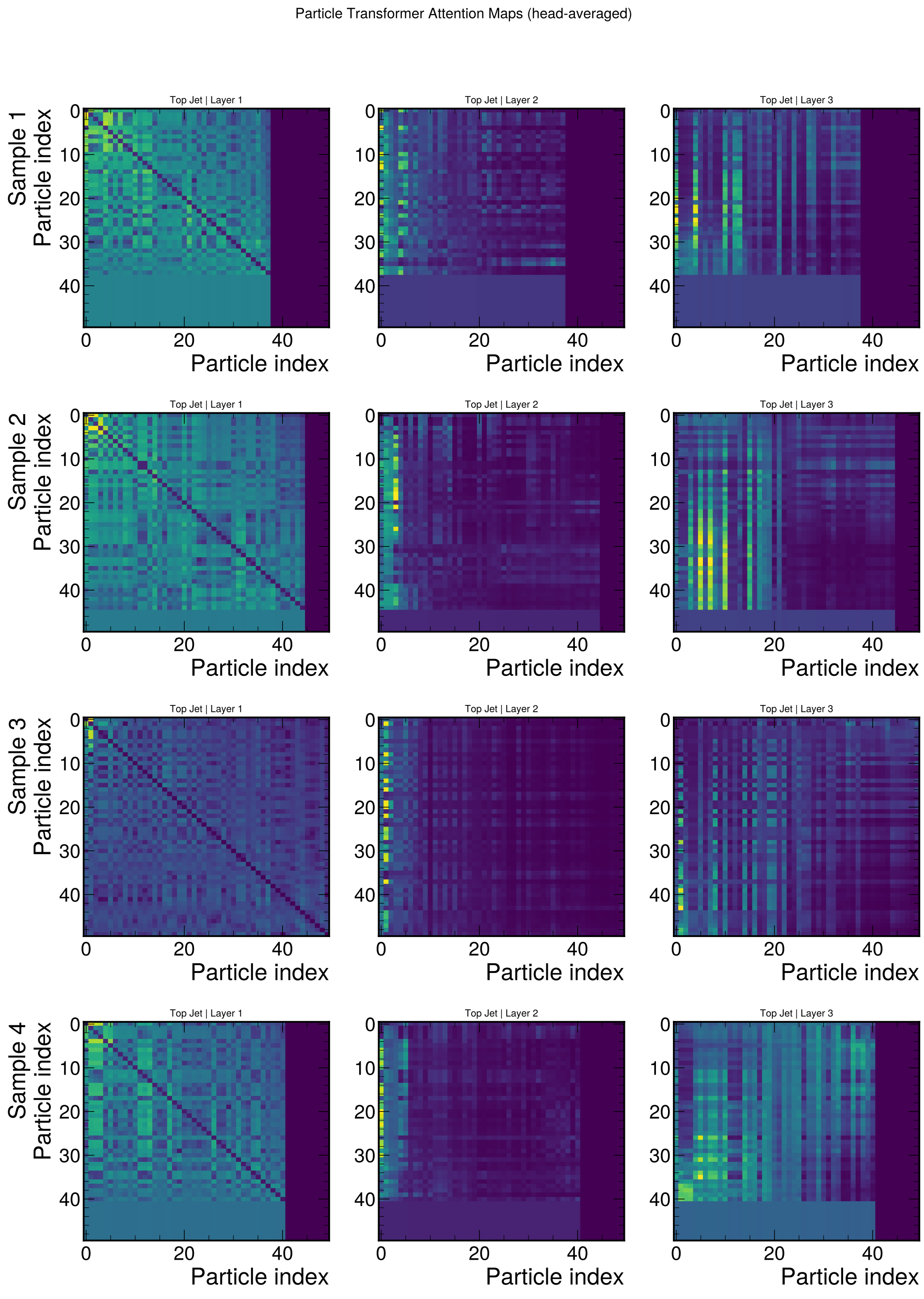

Figure 10: Head-averaged self-attention maps of the Particle Transformer across the encoder layers for representative top jets; padded positions appear as uniform blocks.